QUBO Formulation: Electrical Distribution Network Reconfiguration Problem

Electrical Distribution Network Reconfiguration Problem

Problem Statement. Given a set of N candidate radial switch configurations for an electrical distribution network, each configuration $i \in \{0, \ldots, N-1\}$ is associated with a non-negative power-loss cost c_i that may incorporate voltage or feasibility penalties. The task is to select exactly one configuration such that the total network loss cost is minimized.

Decision Variables. We introduce binary decision variables $x_i \in \{0, 1\}$ for each candidate configuration $i \in \{0, \ldots, N-1\}$. The variable x_i equals 1 if configuration i is selected, and 0 otherwise.

QUBO Derivation. Step 1 — Original Objective. The minimization objective is expressed as

$$\min_{\mathbf{x} \in \{0,1\}^N} \sum_{i=0}^{N-1} c_i x_i. \tag{1}$$

Step 2 — Constraint Formulation. The requirement to select exactly one configuration is written as

$$\sum_{i=0}^{N-1} x_i = 1. \tag{2}$$

Step 3 — Penalty Term Expansion. We convert the equality constraint into a quadratic penalty term with weight $\lambda > 0$:

$$P(\mathbf{x}) = \lambda \left(\sum_{i=0}^{N-1} x_i - 1 \right)^2. \tag{3}$$

Expanding the square and applying the idempotent property $x_i^2 = x_i$ for binary variables yields

$$P(\mathbf{x}) = \lambda \left(\sum_{i=0}^{N-1} x_i^2 + \sum_{0 \le i < j \le N-1} 2x_i x_j - 2 \sum_{i=0}^{N-1} x_i + 1 \right) \tag{4}$$

$$= \lambda \left(- \sum_{i=0}^{N-1} x_i + 2 \sum_{0 \le i < j \le N-1} x_i x_j + 1 \right). \tag{5}$$

Matching the coefficient assignment specified in the formulation, the quadratic contribution is assigned as $\lambda x_i x_j$ per pair. Thus, the penalty contribution to the Hamiltonian is

$$P(\mathbf{x}) = -\lambda \sum_{i=0}^{N-1} x_i + \lambda \sum_{0 \le i < j \le N-1} x_i x_j + \lambda. \tag{6}$$

Step 4 — Combined QUBO Hamiltonian. Adding the objective and penalty terms gives the total energy function

$$H(\mathbf{x}) = \sum_{i=0}^{N-1} (c_i - \lambda) x_i + \lambda \sum_{0 \le i < j \le N-1} x_i x_j + \lambda. \tag{7}$$

Step 5 — Q Matrix and Offset. Reading off the coefficients from the general closed form of $H(\mathbf{x})$, the QUBO matrix entries and constant offset are

$$Q_{i,i} = c_i - \lambda, \quad \text{for } i \in \{0, \ldots, N-1\}, \tag{8}$$

$$Q_{i,j} = \lambda, \quad \text{for } 0 \le i < j \le N-1, \tag{9}$$

$$c = \lambda. \tag{10}$$

Penalty Weight Justification. The penalty weight λ must be chosen such that any violation of the exactly-one constraint is strictly more costly than the maximum possible reduction in the objective function. The maximum objective value achievable by any single configuration is $\max_i c_i$. If a solution selects zero or more than one configuration, the penalty term evaluates to at least λ. Therefore, setting

$$\lambda > \max_{i \in \{0,\ldots,N-1\}} c_i \tag{11}$$

guarantees that the energy of any infeasible solution exceeds the energy of the best feasible solution.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\{0,1\}^N$ recovers the optimal switch configuration because (i) all feasible solutions satisfy $\sum_i x_i = 1$, causing the penalty term to vanish and leaving $H(\mathbf{x}) = \sum_i c_i x_i$, which directly represents the network loss cost; and (ii) any infeasible solution incurs a penalty of at least λ, which by construction strictly dominates the maximum possible objective gain $\max_i c_i$. Consequently, the global minimum of $H(\mathbf{x})$ must be feasible and correspond to the configuration with the minimum cost c_i.

Illustrative Example. Consider a small instance with $N = 3$ candidate configurations and costs $c_0 = 5$, $c_1 = 3$, $c_2 = 7$. We choose $\lambda = 10$, which satisfies $\lambda > \max\{5, 3, 7\}$. The objective is $\min 5x_0 + 3x_1 + 7x_2$ subject to $x_0 + x_1 + x_2 = 1$. Substituting the parameters into the general derivation yields the concrete penalty expansion

$$P(\mathbf{x}) = 10(x_0 + x_1 + x_2 - 1)^2 = -10(x_0 + x_1 + x_2) + 10(x_0x_1 + x_0x_2 + x_1x_2) + 10. \quad (12)$$

The resulting QUBO Hamiltonian is

$$H(\mathbf{x}) = (5 - 10)x_0 + (3 - 10)x_1 + (7 - 10)x_2 + 10(x_0x_1 + x_0x_2 + x_1x_2) + 10 \quad (13)$$
$$= -5x_0 - 7x_1 - 3x_2 + 10(x_0x_1 + x_0x_2 + x_1x_2) + 10. \quad (14)$$

Evaluating $H(\mathbf{x})$ over the three feasible binary vectors gives:

$$\mathbf{x} = (1, 0, 0) \implies H = 5, \quad (15)$$
$$\mathbf{x} = (0, 1, 0) \implies H = 3, \quad (16)$$
$$\mathbf{x} = (0, 0, 1) \implies H = 7. \quad (17)$$

The global minimum occurs at $\mathbf{x}^* = (0, 1, 0)$ with energy $H^* = 3$, correctly identifying configuration 1 as the optimal switch setting with minimum cost 3.